CE2003 Tut 3

Q1 Describe three alternative functions that the LUTs in modern FPGAs can be configured to implement.

Q2 Determine the contents of a 6-input LUT configured as a 4 by 1 multiplexer. Assume the input order is {**sel** [1:0], **d** [3:0]}. It is easiest to look at the patterns from 0 to F for **d** [3:0], then consider which bit is selected for each **sel** pattern. Do not try to write out the 64-row truth table!

Q3 If a 6-input LUT can implement a 6-input function, explain why and how two such LUTs can implement a 7-input function using just the LUTs and a multiplexer.

Q4 Given a 4-input LUT with the contents 16'h84B7, write the function as a sum of minterms, then draw a minimised gate-level circuit and write the corresponding sum of products equation. Explain why this step is not relevant for FPGA implementation.

Q5 The following Verilog **always** block describes a combinational function. Determine the contents of two 3-input LUTs that together implement the 4-input function, showing how they are combined using a multiplexer, and clearly labelling the input signals. Use the input order a, b, c, d.

```
always @ *
begin
   y = 1'b0;
   if (a) begin
      if (b) begin
         if (c ^ d) y = 1'b1
      end
      else
         if (~c) y = 1'b1;
   end else
      if (c) y = 1'b1;
      else
         if (d) y = 1'b1;
end
```

Tutorial 3

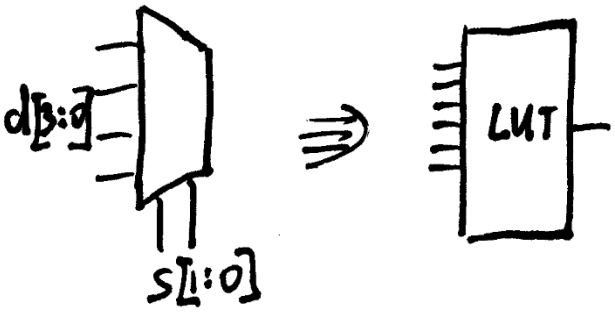
Answer 1

The LUTs/function generators in FPGA logic blocks can implement:

- Shift registers: Shift internal data bits by 1 bit, adding the input bit to the end of the contents, once each clock cycle.
- Memories: Each LUT can implement a 64×1 memory. Multiple LUTs can be combined to implement larger and multi-ported memories.
- Multiplexers: Each LUT can implement a 4 by 1 multiplexer, and multiple LUTs in a CLB can be combined to implement a 16 by 1 mux.

Answer 2

First consider the input pattern for the 4 data bits:



S[1:0]	d[3:0]	LUT
00	⋮	d0
01	⋮	d1
10	11	d2
11	11	d3

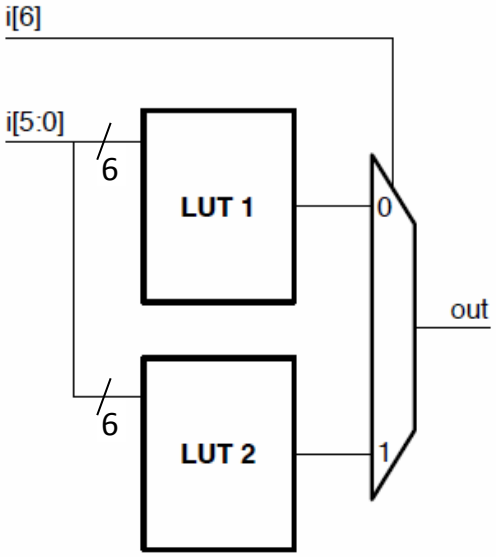
d[3:0]			
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

If sel is 00, we select d[0], for all cases, thus we get 101010101010101010, or 16'hAAAA. Note we go from MSB to LSB (or the reverse of the truth table).
If sel is 01, we select d[1], and get 1100110011001100, or 16'hCCCC.
If sel is 10, we select d[2], and we get 1111000011110000, or 16'hF0F0.
Finally, if sel is 11, we select d[3], and we get 1111111100000000, or 16'hFF00.

Thus, the LUT contents should be 64'hFF00_F0F0_CCCC_AAAA

Answer 3

A 6-input LUT contains sufficient locations for a 6-input truth table, 2^6 or 64 output bits. If we want to implement a 7-input function, we would need 2^7, or 128, entries. We can code the first 64 entries in the first LUT, and the second 64 in the second LUT. In terms of the truth table, this means the entries for an MSB of 0 are in the first LUT, while the inputs for an MSB of 1 are in the second LUT. We can then use the MSB to select between the two outputs using the MSB as a select.

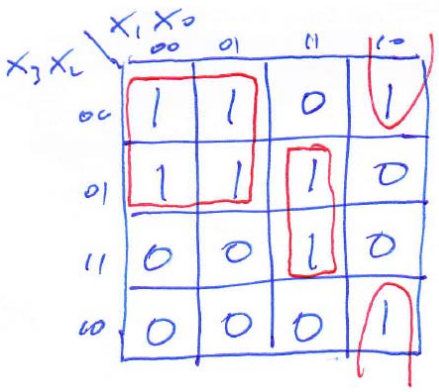


Answer 4:

The LUT contents of 16'h84B7, represent the following sum of minterms:

$F_{x[3:0]} = \sum m(0, 1, 2, 4, 5, 7, 10, 15)$ (LUT contents in reverse order)

We can then draw a K-map:



Simplifying, we get:

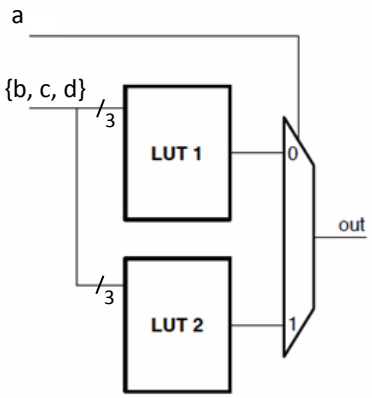
F = (x[3]' x[1]') | (x[2] x[1] x[0]) | (x[2]' x[1] x[0]')

This is not relevant since in the FPGA implementation, the logic is implemented in LUTs, and the propagation delay and area of the logic implementation is a function of the number of inputs and is independent of the specific function being implemented.

Answer 5

Remember the default assignment is at the top, and with the XOR, we can draw the table as:

```
always @ *
begin
    y = 1'b0;
    if (a) begin
        if (b) begin
            if (c ^ d) y = 1'b1;
        end else
            if (~c) y = 1'b1;
    end else
        if (c) y = 1'b1;
        else
            if (d) y = 1'b1;
end
```



a	b	c	d	y
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

Reading up the column, we get 16'h63EE as the LUT contents

Hence, LUT 1 will be 8'hEE and LUT 2 will be 8'h63